

SME ESG & AI SOLUTION



EcoLedger AI

Dual-Ledger Financial P&L & AI Carbon Accounting Engine for

SMEs

Bridging the gap between company profitability (₹ INR) and environmental sustainability (kg CO₂e).

Problem Statement & Audience

Who We Serve

Small and Medium Enterprises (SMEs), supply chain vendors, and manufacturing teams — the economic backbone, currently underserved by legacy ESG software.

The Core Problem

- ❗ **Cost Prohibitive:** Enterprise ESG tools cost \$20,000+/year.
- ❗ **Carbon Blindspot:** SMEs track P&L diligently but lack operational carbon metrics.
- ❗ **Missed Incentives:** Inability to leverage carbon tax rebates & green subsidies.



Proposed Solution & Innovation



Dual-Ledger Sync

Simultaneous, real-time tracking of financial monetary spend (₹ INR) and greenhouse gas emissions (kg CO₂e).



Scope 1–3 Mapping

Automated conversion of raw transaction entries into official GHG Protocol categories without manual carbon audits.



Green Incentives

Direct algorithmic matching of operational expense items against localized carbon tax credit frameworks (e.g., Section 80IA).

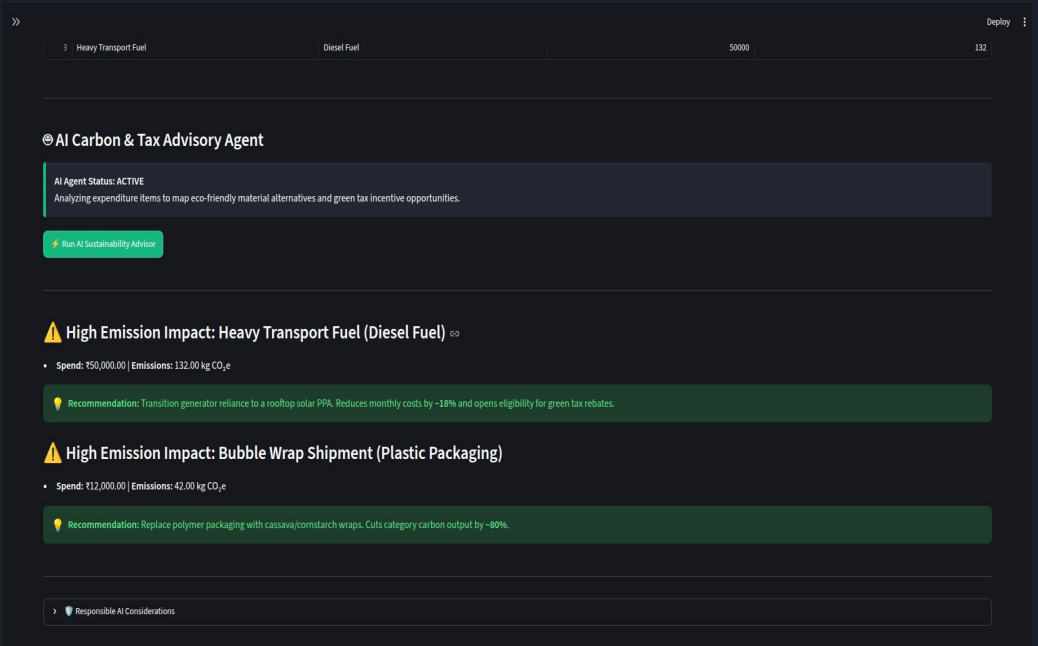
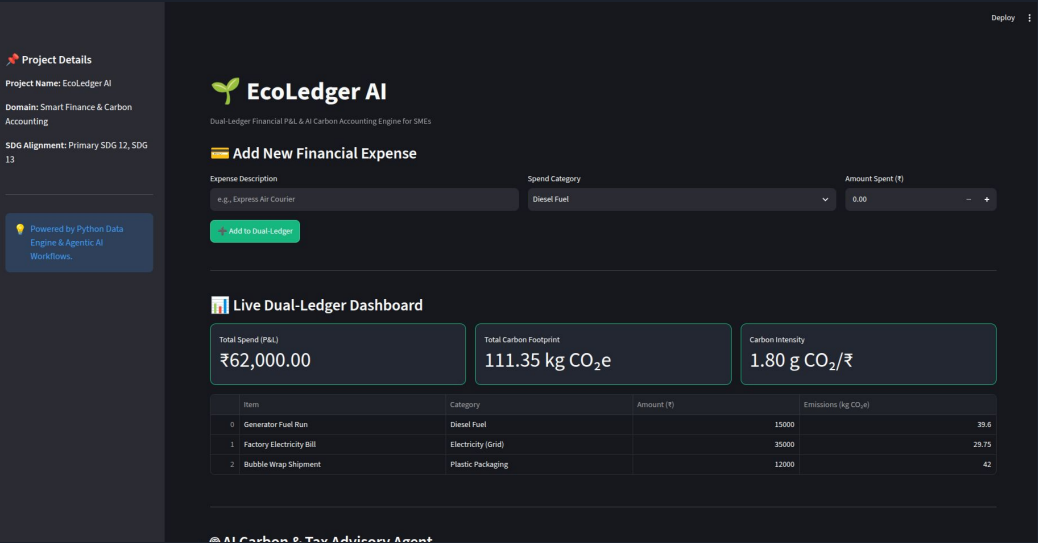
AI & Carbon Metrics Engine

Role of AI in EcoLedger

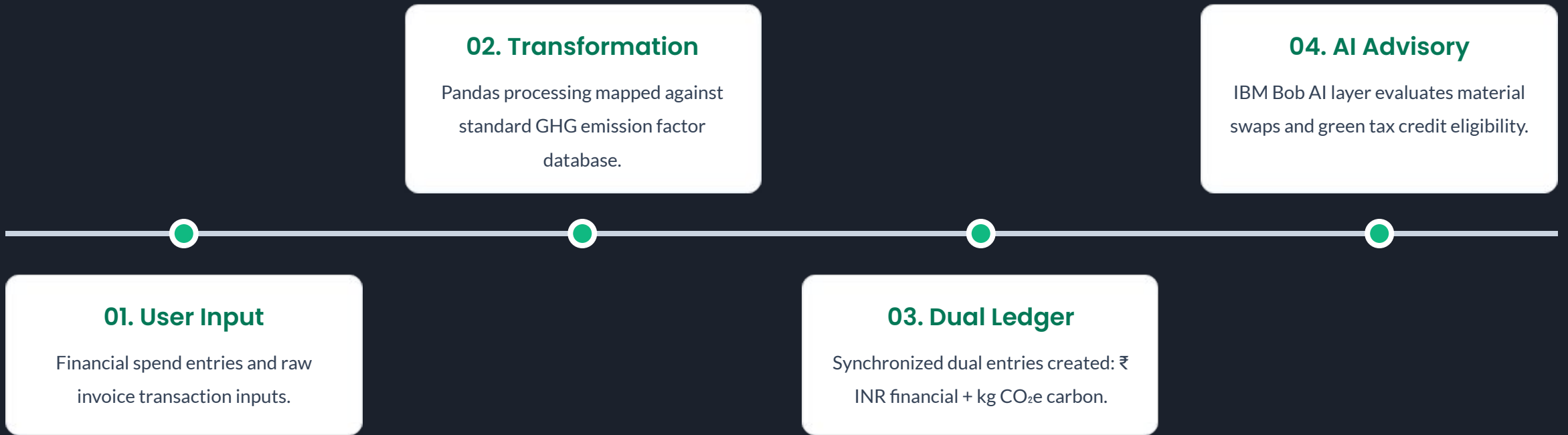
Utilizes algorithmic spend-to-emission conversion via GHG factors alongside a RAG-based LLM engine for eco-friendly supply chain substitutions.

IBM Bob Co–Pilot Integration

IBM Bob optimized the backend data pipeline, structured Streamlit modules, and acts as an active embedded agent flagging high-emission line items and suggesting material swaps (e.g., cassava starch wraps, solar PPAs).



System Architecture Pipeline



UN SDG Alignment

SDG 12

Responsible Consumption & Production

Target 12.6: Encourages companies to adopt sustainable practices and integrate sustainability metrics into financial reporting cycles.

Impact: Directly replaces carbon-intensive packaging and diesel reliance with sustainable alternatives.

SDG 13

Climate Action

Target 13.2: Integrates climate change measures directly into core business operational strategies.

Impact: Empowers SMEs to actively measure, track, and reduce operational Scope 1 and Scope 3 footprints.

Responsible AI Framework



Fairness & Access

Eliminates software licensing barriers by providing low-overhead carbon accounting for SMEs without enterprise budgets.



Transparency

Uses explicit, deterministic GHG mathematical conversion formulas ($\text{Spend} \times \text{Factor} = \text{kg CO}_2\text{e}$) rather than black-box models.



Privacy & Security

Parses only numeric transaction categories and amounts—capturing zero Personally Identifiable Information (PII).

Expected Impact & Future Roadmap

Impact Focus

 **Environmental:** Lowers Scope 1–3 emissions and decreases industrial non-biodegradable waste.

 **Economic:** Protects SMEs against carbon penalties while unlocking government green energy subsidies.

Product Roadmap

01. Accounting APIs: Direct sync with Tally, Zoho Books, & QuickBooks.

02. OCR Invoice Scanning: Instant line-item carbon parsing from receipts.

03. Blockchain Ledger: Immutable ESG compliance verification records.

THE FUTURE OF RESPONSIBLE BUSINESS

Profitability Meets Purpose

EcoLedger AI transforms daily financial decisions into tangible opportunities for environmental stewardship and economic growth.

Drive Profitability • Champion Sustainability • Empower SMEs